

The Witness: full handoff (review, fixes, artifacts, what is left)

Written 2026-09-29 and kept current through the evening of that day (last update after the X12 code work, about 19 hours before the deadline). This is the one document another model needs to pick the project up cold. It records the review that was done, every defect that was found and how it was fixed, every contest artifact that was produced and how to rebuild it, and the work that remains with step-by-step instructions. Where a step depends on knowledge that is not in this repo, a reference is given.

Deadline: **2026-09-30 11:00 ET** (lablab.ai, AssemblyAI Voice Agent Hackathon). Nothing is submitted until the owner clicks submit.

Read in this order: `CLAUDE.md` (hard constraints), this file, `docs/submission/RUNBOOK.md` (the short operational checklist), `docs/PROJECT-STATE.md` (decisions and sourced numbers), `docs/SITUATION-REPORT-2026-09-29.md` (the state before today's work, with the chronology of the audio problem).

0. Where everything stands right now

Area	State on 2026-09-29 evening (UTC)
Branches	<code>main</code> carries everything. The cloud-session branch <code>claude/wizardly-edison-7fk3g9</code> was identical to <code>main</code> and has been deleted. <code>main</code> is the only branch. Twelve PRs (velez2689/the-witness#1 to #12) were merged today.
Checks	<code>npm run typecheck</code> , <code>npm run lint</code> , <code>npx vitest run</code> (19 files, 227 tests), <code>npm run build</code> , <code>npm ci</code> all pass on <code>main</code> .
Deployment	Vercel auto-deploys <code>main</code> to <code>https://the-witness-omega.vercel.app</code> . The build is current when the console shows a "Headphones / Speakers" control beside "Be the Rep (live)".
Contest artifacts	Short description, long description, cover image, 13-slide deck (HTML and PDF), narrated demo video (4:32), judging template: all done and committed. See section 4.
Live call	Proven on 2026-09-29: the owner saved three calls on the current build (<code>docs/assets/calls/</code>), one a full plan to the sign-off. The defects the other two showed (bare name, split badge turn, three-digit badge, "non-covered" untyped, verbatim re-ask) are fixed with tests in <code>tests/domain/live-call-lessons.test.ts</code> . See section 5, step 1.
Commit log	Commits exist on Sep 4, 21, 22, 23, 25, 26, 29 (eleven on the 29th). A commit on Sep 30 before 10:30 ET is still needed.
Secrets	<code>ASSEMBLYAI_API_KEY</code> is set on Vercel, in the owner's local <code>.env</code> , and in the cloud session environment. It is not in git. Rotate it after submission (it was printed into a session log once).

1. What the product is, in one paragraph

A voice agent that rides along on a live phone call a medical biller is making to a health-insurance payer. It records what the payer's representative said as timestamped, quotable statements in an append-only, hash-chained ledger keyed to the claim; detects contradictions against earlier calls on the same claim in real time;

speaks a challenge on the recorded line; reads reference numbers back instead of guessing; refuses to hang up until every required field is captured or refused on record; and emits an appeal packet quoting the transcript. The line that settles design arguments: **we don't predict what the payer will pay, we record what the payer said**. Vocabulary: *Agent* = the human biller, *Rep* = the payer representative, *Witness* = the voice AI. In code: `operator / rep / witness`.

Architecture that judges must hear correctly: two AssemblyAI sockets per call. The Rep's microphone goes to **Streaming STT v3** (`wss://streaming.assemblyai.com/v3/ws`), verbatim, with the claim's identifiers seeded as keyterms. The Witness speaks through the **Voice Agent API** (`wss://agents.assemblyai.com/v1/ws`) as a mouth only: it is never sent audio, calls no tools, and never writes the ledger, by choice. Every evidence-bearing sentence is assembled by our code from ledger rows; the model supplies no facts. Speaker identity is structural (which socket carried the audio), never diarization.

2. The review: what was found and what was done about it

The owner asked for six things: review the situation report; review all code for correctness and for AI craft, emojis and stray symbols; review the visual design and architecture and make it not look AI-generated; fix the failing live call using the saved test calls; write a plan for the contest artifacts; write a judging template. All six were delivered. The findings are listed here so the next model does not re-derive them.

2.1 Situation report corrections (`docs/SITUATION-REPORT-2026-09-29.md`)

The report was accurate on architecture, chronology and the evidence table. Nine corrections were made; the report's section 13 addendum records them. The important ones:

1. **The 1800 ms identifier protection was lost in the two-socket split.** The `turn_detection.min_silence` setting lived on the Voice Agent socket, which since commit `1a3d149` receives no audio. The Streaming socket that actually hears the Rep ran on the server default (about 1.5 s forced end of turn), so a reference number read with a pause became two turns, both too short for the extractor. Fixed by F1 below.
2. **The app's own token rate limit fired before AssemblyAI's.** 10 tokens per 10 minutes per IP, and a call needs 2, so the sixth start inside 10 minutes got a 429. Raised to 20 (F13).
3. **The recap ended the live call before the Rep could answer**, so the sign-off added in `32a1873` was unreachable on the live path. Fixed by F2.
4. **Barge-in could not work as written.** The Voice Agent API has no client event to cancel a reply in flight; `onPartial` flushed the local buffer but the server kept streaming and the rest played after the flush. Fixed by F3 (mute the remainder).
5. **586db95 cannot have fixed the 17:26 and 17:29 failures on Sep 25.** The recordings show the server stopped delivering the greeting after about 2 s; that is upstream of playback, and one of the two attempts had a dead microphone (15 s of digital silence). The diagnosis in section 2.3 below.
6. **The README contradicted the project's own sourced numbers** (appeal overturn claim, denial rate, stale Status table, placeholder clone URL, config variables the code never reads). Rewritten.
7. **The architecture diagram was stale** (showed one Voice Agent socket doing both jobs). The `.d2` source is updated; the SVGs were regenerated with d2 on 2026-09-29.

2.2 Code review findings (F1 to F15), all fixed unless marked

Ranked as they were fixed. File paths are current.

#	Finding	Fix	Where
F15	The call recorder logged no socket events, so a failed call could not be diagnosed.	Added TraceKind events: reply-started (ours/model), reply-done (status), suppressed, transcript-agent, session-error, stt-begin, stt-partial, stt-error, token (kind and HTTP status), socket-closed, interrupted. Routed through LiveEvents.trace so src/ui stays props-only.	src/services/live-call.ts, src/services/call-recorder.ts, src/app/console-host.tsx, tests/services/call-recorder.test.ts
F14	"Save call" fired three downloads at once; Chrome blocked the second and third, so saves came out incomplete and got renamed by hand (the calls README misattributed two files).	One ZIP (call-<stamp>.zip) with stored entries and CRC-32.	src/services/zip-writer.ts (new), console-host.tsx
F1	The Streaming socket had no turn-silence, model pin, voice focus or inactivity settings.	STT_SETTINGS = { speechModel: 'universal-3-5-pro', maxTurnSilenceMs: 2000, minTurnSilenceMs: 400, voiceFocus: 'near-field', inactivityTimeoutSeconds: 120 } sent as query parameters; keyterms and format_turns kept. Not yet confirmed against the live server (section 5, step 6).	src/services/streaming-session.ts, src/services/live-call.ts
F2	Recap hung up before the Rep answered.	pendingClose only for alert_agent and close; recap sets awaitingRecapAnswer; an 8 s silence fallback (RECAP_SILENCE_MS) speaks the close move if the Rep says nothing.	live-call.ts, tests/services/live-call.test.ts
F3	Barge-in left the rest of the reply playing.	VoiceAgentSession.muteCurrent() marks the in-flight reply as not ours until reply.done; onPartial calls stopAudio() then muteCurrent() and traces interrupted.	src/services/voice-agent-session.ts, live-call.ts
F4	The mic gate dropped the Rep's early words on headphones (no echo to gate).	audioSetup: 'headphones' 'speakers' on LiveDeps; gate only for speakers; a two-button segment in the brief panel.	live-call.ts, src/ui/Panels.tsx, src/ui/use-live-call.ts
F5	package-lock.json out of sync; npm ci failed on a fresh clone.	npm install, lockfile committed.	commit b9352c6
F6	The latency figure was engine time only.	Statement offsets come from the server's word timings	streaming-session.ts (TurnTiming), live-call.ts

#	Finding	Fix	Where
		(Turn.words[].start/end); flag latency is measured from the end of the Rep's turn on a sent-audio clock; shown as "flag 1.62 s after end of turn · engine 0.4 ms". Scripted path shows engine-only and says so.	
F7	"Play" buttons re-synthesised text with <code>speechSynthesis</code> , implying a recording existed.	Live statements have "Play recording" (the Rep PCM sliced from the recorder on the transcription clock via <code>repSlice</code>); scripted data says "Read aloud".	<code>call-recorder.ts</code> , <code>src/ui/Panels.tsx</code> (Inspector), <code>console-host.tsx</code> (<code>playSpan</code>)
F8	<code>ClaimTimeline</code> hardcoded <code>'call-06'</code> .	<code>liveCallId</code> prop from <code>Console.tsx</code> .	<code>src/ui/ClaimTimeline.tsx</code>
F9	Stale README facts and diagram.	README rewritten; <code>.d2</code> sources updated; SVGs regenerated with d2 on 2026-09-29.	<code>README.md</code> , <code>docs/assets/architecture.d2</code> , <code>mode-a-call-flow.d2</code>
F10	Dead configuration (<code>ASSEMBLYAI_STT_MODEL</code> , <code>ASSEMBLYAI_AGENT_LLM</code> , <code>ELEVENLABS_API_KEY</code>).	Removed from <code>.env.example</code> and README. Only <code>ASSEMBLYAI_API_KEY</code> and <code>SESSION_MAX_SECONDS</code> are read.	<code>.env.example</code> , <code>README.md</code>
F11	The spike script claimed to send what the service sends but sent rejected shapes.	Payload aligned with <code>session.update</code> in the service.	<code>fixtures/scripts/spike-voice-agent.mjs</code>
F12	A comment said the server sends audio "far faster than real time"; measured roughly real time.	Comment corrected; logic unchanged.	<code>src/services/audio-io.ts</code>
F13	Token limit 10 per 10 minutes = 5 call starts.	<code>MAX_TOKENS_PER_WINDOW = 20</code> . AssemblyAI's 5 streams per minute still bounds abuse.	<code>src/app/api/token/route.ts</code>
Hygiene	Emojis: none found. Em and en dashes, curly quotes and ellipses replaced with plain punctuation in UI strings and docs. Middle dots kept as the console's separator. <code>docs/STYLE.md</code> added.		

Things deliberately kept: long narrative comments (accurate and traceable), the `onUserTurn` tripwire on the voice socket, `resume` , `editDistance` , `addDays` .

2.3 What the saved calls actually showed (the live-call diagnosis)

Method: every `call-*.json` was dumped and a 50 ms RMS envelope computed over every WAV. The witness WAV is the audio *as it arrived from the server* and passed the ownership filter, so its length is what the server

delivered, not call time. Details and the corrected file index are in `docs/assets/calls/README.md`.

Call (UTC, Sep 25)	Build	Witness audio delivered	Mic track	Meaning
20:24:56	one socket	66 s, healthy	149.8 s of speech	Plan ran to the end but the badge was split mid-utterance ("Badge number 2."), so identity was never captured and the call ended on the hand-off, not the sign-off. Real evidence for F1.
20:30:07	one socket	76 s, healthy	208 s	A read-back requested at 144.0 s did not play until 176.1 s, right after the next Rep turn: the server sat on <code>reply.create</code> for 32 s. Same "server not delivering" family.
21:26:01	two sockets, before <code>586db95</code>	1.9 s	digital silence for all 15.3 s	Dead microphone on the owner's machine, yet the greeting still stopped arriving after 1.9 s.
21:29:34	two sockets, before <code>586db95</code>	2.3 s	0.85 s of speech at 10.4 s	The server delivered about 2 s of greeting then nothing; the <code>identify</code> line was queued but no audio ever arrived.

Conclusion: the failure is upstream of playback (server stopped streaming, or a second `reply.started` made the ownership filter drop the rest). The old timeline could not tell these apart. After F15 every saved call names its own cause. One attempt also had a dead input device, which no code fixes; hence the environmental checklist in section 5, step 2.

2.3b Added later on 2026-09-29: X12 codes recognised on the call

The owner supplied the X12 Remittance Advice Remark Code and Claim Status Code lists.

`fixtures/scripts/build-x12-codes.mjs` generates `src/domain/x12-codes.ts` with identifiers, an active flag and a bucket word we assign (1216 remark codes, 796 claim status codes; no description text, same rule as CPT). `src/lib/spoken-codes.ts` hears the spoken and written forms ("N thirty", "M A one thirty", "status code twenty-one", "CO fifty", "co-50"); the extractor files them as `remark_code`, `claim_status_code` or `denial_code` rows, confirmed when on the list, otherwise unconfirmed and read back like a reference number (`call-session.ts` confirms any read-back category; `call-plan.ts` step 6b). A remark code fills the gate item that could only be refused before; a CARC satisfies the denial-reason objective; the engine raises a value conflict when a later call gives a different code; the recap, close-out and claim update sheet carry the codes. Tests: `tests/domain/x12-codes.test.ts`. To refresh the lists: `node fixtures/scripts/build-x12-codes.mjs` (fetches `x12.org`; or pass two saved pages as files).

2.4 Visual and architecture review (V1 to V10)

Verdict: the layout passes the structural test (five bands from named instruments: recorder transport, ATC strip bay on a date axis, multitrack edit view, CMS-1500 on paper, exit checklist; unequal panel widths; hue-biased palette; system fonts; serif italic quotes; confidence as state). The tells that remained were fixed:

#	Tell	Fix
V1	Accent rail on rounded, shadowed cards	Rail kept; radius and shadow dropped on cue, wrong-claim and packet-conflict; only the flag banner keeps elevation
V2	Pills (link chip, "ON NOW", held list)	Rectangular region label on the connector; accent text with the pulse dot; plain comma list

#	Tell	Fix
V3	Too many uppercase letterspaced micro-labels	Kept three (REC, the contradiction kind, CMS-1500 box labels); rest sentence case
V4	Big-number stat strip at the top	Deleted (<code>StatStrip.tsx</code>); the brief carries one computed sentence; contradictions and refusals as words in the transport bar
V5	One radius and shadow everywhere	Timeline and roster cards 3 px; buttons keep 8 px
V6	"Nothing yet" empty states	At idle the historical link (call 02 to call 05, "on record") is drawn in grey, so the still frame shows the product before Start
V7	The link (the thesis) was the thinnest mark	2.5 px, endpoint cards elevated, region label readable at thumbnail size
V8	Edit view had no waveform	Real amplitude envelope inside each live Rep block from the recorder (<code>repPeaks</code>); scripted blocks stay solid
V9	Stale GIF and diagrams	GIF, cover and screenshots regenerated (<code>fixtures/scripts/capture-console.mjs</code>); SVG diagrams regenerated with d2 on 2026-09-29
V10	Packet section headers uppercase mono red	Bold sans, sentence case

Architecture of the product: sound. The one honest boundary to state to judges: statement extraction is rule-based on the demo vocabulary (`src/domain/extractor.ts`); a differently phrased denial is captured as a verbatim quote, not a typed fact. Deck slide 8 says this.

3. What was done today, commit by commit

All on `main` via merged PRs. Conventional commits; each has a real body.

Commit	PR	What
<code>b9352c6</code> chore	#1	Sync <code>package-lock.json</code> so <code>npm ci</code> works
<code>24a5968</code> docs	#1	README, config and diagram source brought up to the two-socket build; plain punctuation
<code>1e92ff7</code> fix(live)	#1	F15, F14, F2, F4, F3, F1 with tests
<code>87b3f65</code> feat(ui)	#1	V1 to V7, V10; the record shown before the call starts
<code>8382143</code> docs	#1	Submission copy, deck, judging template, runbook
<code>8f24d57</code> fix(ui)	#1	F8, F7 (labels), F13
<code>40d2733</code> docs	#1	The twelve steps at the top of the runbook
<code>f592a6d</code> feat(live)	#2	F6, F7 (recorded playback), V8
<code>33f933d</code> docs(deck)	#3	Competitor rows verified against vendor pages, sources on the slide, AKASA contrast
<code>1ba1924</code> chore	#3	Footage capture script

Commit	PR	What
c954b12 docs	#4	Draft picture track (later replaced)
badb2c6 docs	#5	Narration script; render, capture and mux scripts
5e07ea3 docs	#6	Narration cut to 689 words to fit the length rule
353d465 docs	#6	Narrated demo video (4:16), per-line audio, slide 8 and long-description paragraph filled
db0c0a9 docs	#7	2025 CAQH Index cited alongside the 2024 per-transaction figure
6702515 docs	#8	This handoff document
afbfb487 docs	#9	PDF export of the handoff
61eb96b docs	#10	Demo video re-cut at 4:32 with Whisper mode, citations readable on screen, telephony removed; Mode B renamed Whisper mode everywhere
4d0dfb5 docs	(owner)	The three live calls of 2026-09-29 saved as zips
fca9cd2 fix(live)	#11	What the first three live calls taught: bare name, split badge turn, three-digit badge, "non-covered", no verbatim re-ask, muted trace label
173061f feat(domain)	#12	X12 remark, claim status and CARC codes recognised when read out; identifiers only

Total across the day: about 130 files changed since the owner's last commit before the review, including the generated code table, the narration audio and the video.

4. The contest artifacts: what exists and how to rebuild each one

All text copy lives in `docs/submission/`, all media in `docs/assets/`. Standing rules for every artifact: no transcription accuracy percentage; no CPT descriptor text; no PHI (all demo data is synthetic and says so); no rep reputation scoring; never call footage live unless it is; every number carries its source and data year; plain punctuation; no emoji.

4.1 Short description

`docs/submission/short-description.txt`, 252 characters (limit 255). Check: `python3 -c "print(len(open('docs/submission/short-description.txt').read()).strip()))"`.

4.2 Long description

`docs/submission/long-description.md`. Five paragraphs: the problem with sourced numbers; the line; what it does; why AssemblyAI; what the video shows (filled: the scripted Rep on the real deployed console, narrated by an AssemblyAI voice agent). Check: `wc -w` at or above 100, and the prohibited-claim grep in section 6.

4.3 Cover image

`docs/assets/cover-16x9.png`, 3840x2160 (downscale to 1920x1080 on upload if the form wants it). Left third: title, the line, the hackathon name. Right: the console at the moment of the value conflict. Rebuild: start the production server (section 7), then `PLAYWRIGHT_CORE=<path> node fixtures/scripts/capture-console.mjs` (also regenerates `console-run.gif`, `console-flag.jpg`, `packet.jpg`).

4.4 Deck

`docs/submission/deck.html` (one `<section>` per slide, the console's own CSS tokens) printed to `docs/submission/deck.pdf` by `PLAYWRIGHT_CORE=<path> node fixtures/scripts/render-deck.mjs`. Thirteen slides: 1 title; 2 the problem (the refusal quote, 25 min / \$13.80, 19 percent denied, under 1 percent appealed); 3 what vanishes (call 03, hold time to scale); 4 the product in one frame; 5 how it works (two sockets, mouth not hand); 6 the evidence model; 7 why AssemblyAI; 8 what is live, replayed, scripted, roadmap; 9 market; 10 revenue; 11 competitive analysis (Infinitus, Outbound AI, Waystar, Experian Health, AKASA; verified against vendor pages 2026-09-29, sources printed on the slide); 12 roadmap and the ethics line; 13 repo, URL, how to run. Check page count: `python3 -c "import re;print(len(re.findall(rb'/Type\s*/Page[^\s]*',open('docs/submission/deck.pdf','rb').read())))"` prints 13.

4.5 Demo video

`docs/assets/demo-narrated.mp4` : 4:32, 1920x1080, H.264 video with AAC mono audio at 24 kHz. Rule: 3:00 to 4:59. Narrated by an AssemblyAI voice agent, rendered line by line through the same Voice Agent API the product uses (the first line is the session greeting, every other line is `reply.create` with "Say exactly the following", voice `jean`, no `11m` key). It opens "Hi. This is an AssemblyAI voice agent" and closes with the instructions for running the live call. The picture track is the deck for the framing beats and the real console stepped line by line for the call (scripted Rep, stated on screen and on slide 8): the Witness-speaks call first, then three beats of Whisper mode where the biller speaks and the Witness whispers.

Pipeline, in order (all three scripts are in `fixtures/scripts/`, all documented at the top of each file):

```
# 1. Speak the 40 lines of docs/submission/narration.md; writes docs/assets/narration/NN.wav +
index.json
node --env-file=.env fixtures/scripts/render-narration.mjs docs/assets/narration
# 2. Picture track, one beat per line, held for the length of its audio; needs the production
server on :3000
PLAYWRIGHT_CORE=<path-to-playwright-core> node fixtures/scripts/capture-footage.mjs <footageDir>
http://localhost:3000 docs/assets/narration
# 3. Lay the audio at each beat's start and encode
node fixtures/scripts/mux-narration.mjs <footageDir> docs/assets/narration docs/assets/demo-
narrated.mp4
# 4. Check
ffmpeg -i docs/assets/demo-narrated.mp4 2>&1 | grep -E 'Duration|Stream'
```

Only step 1 needs the key and network. If a single line sounds wrong, edit that line in `narration.md`, re-run step 1 (it re-renders all lines; about two minutes and one Voice Agent session), then steps 2 and 3. The narration is 745 words; the voice speaks at roughly 2.9 words per second, and the capture adds 0.6 s per beat plus a 4 s end card, so keep the script under about 750 words to stay under 4:59.

4.6 Judging template

`docs/JUDGING-TEMPLATE.md` : the four lablab criteria (Application of Technology, Presentation, Business Value, Originality), equal weight, 1 to 5 band descriptors, evidence checks, red flags that cap a score, and the output format. Use it to self-score before submitting.

4.7 Runbook

`docs/submission/RUNBOOK.md` : the twelve steps to submission, the state table, the live-call proof steps, the video recipe, the deck checks, and the B6 submission field list.

5. What is left, with step-by-step instructions

Ordered by value per hour. Steps 1 to 5 need the owner's machine (a microphone, headphones, Chrome). Steps 6 to 10 can be done by a model with the key and network.

Step 1. Prove the live call on production (about 30 minutes). Done once on 2026-09-29; repeat on the build that carries the fixes

The first three saved calls are analysed in `docs/assets/calls/README.md`. Run one more call on the fixed build to confirm the bare-name and split-badge cases; the steps are unchanged.

1. Open `https://the-witness-omega.vercel.app` in Chrome. Confirm the build: a "Headphones / Speakers" control sits beside "Be the Rep (live)". If it is missing, Vercel has not finished deploying `main`; wait, or check the deployment list in the Vercel dashboard.
2. Fix the input device first. On Sep 25 one attempt was 15 s of microphone silence. Windows Settings, Sound, Input: select the headset microphone, unmute, turn audio enhancements off. In Chrome's address bar mic permission pick the same device. Speak and watch the input meter move.
3. One tab, headphones on, DevTools Console open (F12), extensions off, no VPN, no screen recorder that captures system audio (on Sep 23 one delayed the greeting by 81 s). Wait two minutes since any earlier attempt: the account allows 5 new streams per minute and one call opens two.
4. Click "Be the Rep (live)", choose "Headphones", click "Start live call". The Witness greets you. Play the Rep from `fixtures/scripts/rep-bank-call-06.ts`: give a name and a four-digit badge; say the claim denied for timely filing; when asked which diagnosis code, say it does not specify; when asked for the reference number, say "eight K two J, nine eight eight"; confirm the read-back; answer the recap; say goodbye. Let it reach "have a good one".
5. Click "Save call". One `call-<stamp>.zip` downloads with `call-*.json` (timeline), `witness-*.wav` and `rep-*.wav`.
6. If anything went wrong, open `call-*.json` and read the events in this order: both `token` events say `ok`; `stt-begin` arrived; `reply-started` with `ours` for the greeting and its `reply-done` with `completed`. Audio that stops with no `reply-done` means the server stalled. A `reply-started` marked `model` means the ownership filter dropped the rest (report it; the split should have removed it). A `reply-done` marked `muted` is our own line cut short by a barge-in and is expected. Any `session-error` text names the cause; anything about limits means wait two minutes. `stt-partial` entries show what the transcriber heard in real time.
7. Copy the zip into `docs/assets/calls/`, add a row to `docs/assets/calls/README.md` with the build hash (`git rev-parse --short HEAD`), commit as `docs: saved live call on <hash>`, push to `main`.

Done when one saved call reaches the sign-off with: greeting complete, identity captured, a challenge spoken, read-back spoken, sign-off heard, no `playback-idle` inside a line, and the transport bar shows a "flag N s after end of turn" figure under 2 s.

Step 2. If the live call fails: what to change

- Greeting never arrives, `reply-started` missing: the Voice Agent socket did not get `session.ready` or the update was rejected. Check `session-error`. Do not add fields to `session.update` (CLAUDE.md constraint 4).

- Greeting starts then stops, no `reply-done` : server stall. Try once more after two minutes. If it repeats, save both zips; this is the server behaviour seen on Sep 25 and the evidence is what matters.
- Rep turns split ("Badge number 2."): the STT socket rejected the turn-silence parameters. Run step 6 to see which; remove it from `STT_SETTINGS` in `src/services/live-call.ts`.
- Nothing transcribed, `stt-begin` present, no `stt-partial` : microphone. Step 1.2 again.
- "Too many sessions started": the app's own limit (20 tokens per 10 minutes per IP). Wait or restart the dev server.

Step 3. If a clean live call exists: put it in the video (optional, 1 hour)

The narrated video already says the call is scripted. To swap in the live call, lay `witness-*.wav` and `rep-*.wav` from the zip at their offsets (`atMs` of the first `playback-started` and the first `rep-turn` in `call-*.json`) under a screen recording of that call made with system-audio capture OFF, cut it into the narrated video between beats 4 and 25, keep the caption "live call on the deployed build", then change deck slide 8's "Demo video call" row and the long description's last paragraph to say so. Re-render the deck PDF. Length must stay under 4:59. If this cannot be finished cleanly, leave the scripted version; an honest scripted demo scores better than a broken live one.

Step 3b. One more live call on the build that carries the fixes and the codes (owner, 10 minutes)

Same steps as step 1. Answer the identity question with only your surname, then "badge number", then the digits. Give "CO fifty" as the denial reason and "N thirty" when asked for the remark code; when the Witness reads "C O, five zero" back, say yes. Expected in the saved timeline: a `rep_identity` row after the digits, a `denial_code` row confirmed by a second row after your yes, a `remark_code` row, and the recap naming both codes. Push the zip as before.

Step 4. Commit on Sep 30 before 10:30 ET

The commit log is judged. Any real change counts: the saved call from step 1, the regenerated SVG from step 8, a README line. Push to `main`.

Step 5. Submit on lablab.ai (owner only)

Fields: title "The Witness: payer call memory"; short description (4.1); long description (4.2); tags AssemblyAI Streaming STT, AssemblyAI Voice Agent API, Next.js, Vercel, healthcare, voice agent; cover (4.3); video (4.5, MP4); slides (4.4, PDF); GitHub <https://github.com/velez2689/the-witness> ; application <https://the-witness-omega.vercel.app> . Pre-submit gate:

- The live URL opens in a fresh incognito window with no auth wall and the scripted call runs end to end (click "Start call").
- `git clone` then `npm ci && npm run build` succeed on a clean machine.
- `LICENSE` is MIT; the README top has the URL.
- The prohibited-claim grep (section 6) returns nothing.
- The owner has watched the video end to end.

Step 6. Confirm the transcription parameters (DONE 2026-09-29 with `fixtures/scripts/spike-streaming.mjs` : no parameter was rejected, and a reference split by a 1.2 s pause arrives as one turn with these settings and as two turns without them)

`STT_SETTINGS` (F1) was added from the AssemblyAI docs without a live spike, against the project's own rule. Write `fixtures/scripts/spike-streaming.mjs` modelled on `spike-voice-agent.mjs` : mint a token from the app's route or directly (`GET https://streaming.assemblyai.com/v3/token?expires_in_seconds=60` with

Authorization: <key>), connect to `wss://streaming.assemblyai.com/v3/ws` with the exact query string `StreamingSession.connect` builds (`sample_rate` , `speech_model` , `max_turn_silence` , `min_turn_silence` , `voice_focus` , `inactivity_timeout` , `keyterms_prompt` , `format_turns`), stream a WAV of someone saying "Eight K two J, nine eight eight" with a 1.2 s pause (render it with `render-narration.mjs` on a one-line file, resample 24 kHz to 16 kHz), and print every `Begin` , `Turn` (with `words`) and `Error` . Done when the reference arrives as one turn. If an `Error` names a parameter, remove it from `STT_SETTINGS` , run the checks, commit `fix(live): drop <param>, rejected by the streaming server` .

Step 7. Verify every narration line was spoken as written (DONE 2026-09-29 with `fixtures/scripts/check-narration.mjs` and an Antigravity listen; the voice dropped small words in lines 12, 18, 25 and 26, which are being re-rendered)

The voice agent may occasionally rephrase. Transcribe `docs/assets/narration/*.wav` (any STT; AssemblyAI's Streaming socket from step 6 works, 24 kHz mono PCM16, skip the 44-byte header) and diff against `narration.md` . Re-render any line that differs (section 4.5).

Step 8. Regenerate the architecture SVGs (DONE 2026-09-29 with d2 v0.9.0)

`d2` is not installable in the cloud sandbox (the network policy blocks the download). Install from <https://d2lang.com/tour/install> , then:

```
d2 docs/assets/architecture.d2 docs/assets/d2-architecture.svg
d2 docs/assets/mode-a-call-flow.d2 docs/assets/d2-mode-a-call-flow.svg
```

Commit `docs: regenerate architecture diagrams from the updated d2 sources` . The README embeds `d2-architecture.svg` ; check it renders on GitHub.

Step 9. Rotate the API key after submission (owner, 2 minutes)

AssemblyAI dashboard, API keys, create a new key, update Vercel's environment variable and the local `.env` , delete the old key. Reason: a check in a cloud session printed the key's value into that session's log.

Step 10. Roadmap items, only if the deadline allows (do not start before steps 1 to 5)

- `agent_context` on the transcription socket: after each Witness line, send `UpdateConfiguration` with the line's text; the docs say it improves recognition of spelled-out identifiers in the next user turn. Spike first.
- **Live Mode B** on a second Streaming session; **Payer Inconsistency Index** at payer level only.

6. Verification commands (run all before any push)

```
npm run typecheck && npm run lint && npx vitest run && npm run build
npm ci # on a fresh clone; proves the lockfile
grep -rniE '[0-9]+ ?% ?(accura|wer)' docs/submission docs/assets README.md # must print nothing
grep -rn 'CAQH' README.md docs/submission docs/PROJECT-STATE.md src | grep -v -E '2023 data|2024
data|2024 edition|2025 edition' # every hit must carry a vintage
```

Never report a check that was not run. If a check fails, say so with the output.

7. Environment and tooling notes

Tooling on the owner's Windows machine

- Earlier work on 2026-09-29 ran in a Claude Code cloud session. That environment is no longer needed and nothing in the repo depends on it.
- ffmpeg is bundled in `ffmpeg-static` (installed under the witness-call-review skill); the scripts also accept `FFMPEG=<path>`. `ffmpeg-static` v0.9.0 is on the PATH. `playwright-core` with Chromium is used for `capture-footage.mjs`, `capture-console.mjs` and `render-deck.mjs` through `PLAYWRIGHT_CORE=<path>`.
- Listening to audio is done by Antigravity (agy , Gemini), never by Claude. Give it an absolute path and forbid shell commands in the prompt; do not grant `command(*)`.
- Production server: `npm run build && npx next start -p 3000`. Never `kill -f "next start"` from inside a session, it kills the session's shell.
- The key is read from the environment as `ASSEMBLYAI_API_KEY`, locally from `.env` (`node --env-file=.env ...`). Never paste it into chat, a commit or a comment.

On the owner's computer

```
git clone https://github.com/velez2689/the-witness && cd the-witness
cp .env.example .env          # set ASSEMBLYAI_API_KEY
npm ci && npm run build && npx next start -p 3000
npm i -D playwright-core      # for the capture and render scripts
```

Every artifact in this repo can be rebuilt with the commands in section 4.

8. References

Verified in this session unless marked.

- AssemblyAI Streaming STT v3: endpoint `wss://streaming.assemblyai.com/v3/ws`; parameters `sample_rate`, `speech_model`, `max_turn_silence` (default 1536 ms; raise it when identifiers split across turns), `min_turn_silence`, `voice_focus`, `inactivity_timeout`, `keyterms_prompt`, `format_turns`; Turn messages carry `words[]` with `start / end` in ms. Docs entry point: <https://www.assemblyai.com/docs> (search "Universal-Streaming").
- AssemblyAI Voice Agent API: endpoint `wss://agents.assemblyai.com/v1/ws`, token `https://agents.assemblyai.com/v1/token`; client events `input.audio`, `session.update`, `session.resume`, `session.end`, `tool.result`, `reply.create`, `conversation.message`; there is no reply-cancel event; `session.update` rejects any `llm` key and a rejected update runs the call on defaults with no greeting (verified live 2026-09-22). Docs entry point: <https://www.assemblyai.com/docs> (search "Voice Agent API").
- Account limits: 5 new streams per minute; billing on socket-open duration.
- lablab.ai judging: four equal criteria; video 3:00 to 4:59; competitive-analysis slide required; commit spread across the window is evaluated. Event page on <https://lablab.ai>.
- CAQH Index 2024 edition (2023 data): 25 min / \$13.80 per manual phone claim-status inquiry. CAQH Index 2025 edition (2024 data, published 2026-02-19): public Executive Report at https://index.caqh.org/hubfs/2025-26_Index_Dashboard/AskTheIndex_KnowledgeBase/2025%20CAQH%20Index%20Report%20Executive%20Report.pdf; per-transaction figures are behind the paid Index Pro. See `docs/PROJECT-STATE.md` section 9 for every

other number (KFF March 2026, Premier February 2025, HFMA/Guidehouse April 2026, GAO-06-710, DOL/EBSA letter of 2021-06-14).

- d2 diagrams: `https://d2lang.com`.
- Repo layout contract: `.repo-layout.yml` ; prose in `docs/ARCHITECTURE.md` . Run its placement procedure before creating any file.
- Style: `docs/STYLE.md` (ASCII punctuation, middle dot as separator, no emoji, no percentage for confidence).

9. Glossary

- **Agent / Rep / Witness**: the human biller, the payer representative, the voice AI.
- **Mode A**: the Witness speaks to the Rep. **Mode B**: Whisper mode, the Witness whispers to the Agent (scripted only today).
- **Scripted Rep / "Start call"**: the built-in simulation of call 06 on claim A-4471-08; no network, no microphone. **"Be the Rep (live)"**: a person plays the Rep into the microphone; two AssemblyAI sessions open on click.
- **Flag**: a contradiction detected against an earlier statement; five kinds (value conflict and four others in `src/domain`). **Refusal**: a category named without a value, recorded as a row.
- **Hang-up gate**: the exit condition, every required field captured or refused on record.
- **Read-back**: the Witness repeats an unconfirmed identifier and asks for confirmation; the confirmed value is a new row, the original is never overwritten.
- **Ledger**: append-only, hash-chained statement rows keyed to the claim; the packet's "Verify integrity" recomputes the chain.
- **Save call**: one zip with the timeline JSON and both audio tracks, from the in-browser recorder.